

Collision/Obstacle Avoidance Dynamic Formation Reconfiguration of High-Order Nonlinear Multi-Agent Systems

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Abstract—Multi-Agent formation involves the motion conflicts of large numbers of physical systems, which leads to significant safety challenges in terms of collision/obstacle avoidance. This paper studies the problem of collision/obstacle avoidance formation tracking for a class of high-order strict-feedback nonlinear multi-agent systems and presents a dynamic formation reconfiguration method. First, a smooth and analytic reference trajectory is constructed, which consists of a convex combination of preset trajectory and delayed target trajectory with its coefficient adaptively adjusted by two barrier functions to avoid collisions. Then, a distributed backstepping control scheme with high-order filtering for barrier functions is proposed. Resultantly, all agents adaptively reconfigure their formation shape within the barrier function's coverage to avoid collisions with each other and obstacles, while quickly forming a pre-defined shape of formation following the target trajectory outside the barrier function's coverage. Simulation results are given to illustrate the effectiveness of the proposed scheme.

Index Terms—Formation reconfiguration, collision/obstacle avoidance, multi-agent systems, barrier functions.

I. INTRODUCTION

In multi-agent formation control, collision/obstacle avoidance as an important safety objective has been widely investigated, and many important collision/obstacle avoidance methods have recently been developed, such as barrier-Lyapunov function (or artificial potential field) method [1]–[10], optimization and game [19]–[22] and geometric method [23]–[26]. Due to the natural combination of potential (barrier) functions with Lyapunov functions, the Barrier-Lyapunov function (BLF) approach has been widely applied to multi-agent coordination. Its main idea is to construct a positive-definite function which is a combination of a quadratic function for coordination control and a barrier function for collision avoidance, and then develop an integrated controller comprising of coordination term and collision avoidance term. The works in [1]–[5] study the collision and/or obstacle avoidance formation control problem of multiple single/double-integrator models. More general nonlinear uncertain Lagrangian dynamics have been

studied in [6]–[9], however, which only focus on leadless formation control but not to consider the tracking task.

In fact, the existing collisions-free formation tracking control methods based on the BLF usually generate the dynamic mismatching [17], [18], [23]. That is, the coordination term and collision avoidance term are offset each other such that the system potentially falls in some unexpected equilibriums, which makes the formation tracking task impossible to be completed. Although many algorithmic solutions have been proposed to solve this problem [10]–[16], an analytical and smooth solution still remains open, which is often required in the tracking control. This is the main technical challenge to address the issue of collision avoidance of formation tracking control based on the BLF approach.

Besides of the BLF approach, there are other collision avoidance methods. In [23], [24], the safety barrier certificate is designed on double-integrator multirobot systems, where the safety barrier controller that approximates the normal coordinated control laws in a least-square sense is derived by solving a quadratic program. The work in [19]–[21] recently gives a differential game solution to the problem of multi-agent collision avoidance, where each agent reaches a target position while avoiding collisions with obstacles and other agents. However, these collision avoidance methods are only suitable to simple single/double-integrator agent's model.

Motivated by these, this paper studies the problem of collision/obstacle avoidance formation tracking for a class of high-order strict-feedback nonlinear multi-agent systems (MAS) and presents a backstepping-based dynamic formation reconfiguration method. The overall reconfiguration architecture consists of two steps: i) trajectory planning: generate a collisions-free trajectory; ii) trajectory tracking: take the planned trajectory as reference and construct a distributed tracking control scheme. The technical contributions of this paper include:

- Due to the n -order time derivatives of the reference

trajectory required in the backstepping framework, the algorithmic methods [10]–[16] are not efficient. To this end, we derive a smooth and analytic reference trajectory in terms of a convex combination of preset trajectory and delayed target trajectory with its coefficient adaptively adjusted by two barrier functions to avoid collisions.

- To avoid using neighboring agents' high-order state variables in repeatedly differentiating reference trajectory, high-order filters for the barrier functions are constructed.
- With the proposed control scheme, all the agents adaptively reconfigure their formation shape within the barrier function's coverage to avoid collisions with each other and obstacles, while quickly forming a pre-defined shape of formation following the target trajectory outside the barrier function's coverage.

Notations: $\mathbb{N}$ and $\mathbb{R}$ denote the sets of natural and real numbers, respectively. The subset of $\mathbb{N}$, $\{1, \dots, N\}$ is denoted by $[N]$. $\mathbb{R}^n$ represents the n -dimensional Euclidean space. Given a positive-definite matrix $X \in \mathbb{R}^{n \times n}$ and a vector $x \in \mathbb{R}^n$, we define $\|x\|_X = \sqrt{x^T X x}$.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Problem Formulation

Consider N heterogeneous agents moving on the n -dimensional Euclidean space. The dynamics of agent i is modeled as following uncertain nonlinear strict-feedback form:

$$\begin{cases} \dot{x}_{i,1} = x_{i,2} + \theta_i^T \varphi_{i,1}(x_{i,1}) \\ \dot{x}_{i,2} = x_{i,3} + \theta_i^T \varphi_{i,2}(x_{i,2}) \\ \vdots \\ \dot{x}_{i,n} = u_i + \theta_i^T \varphi_{i,n}(x_i) \\ y_i = x_{i,1} \end{cases} \quad (1)$$

where $\bar{x}_{i,j} = [x_{i,1}, \dots, x_{i,j}]^T \in \mathbb{R}^{j \times n}$, $\bar{x}_i = \bar{x}_{i,n_i} \in \mathbb{R}^{n_i \times n}$ is the measurable system state, $u_i, y_i \in \mathbb{R}^n$ are control input and system output, respectively, $\theta_i \in \mathbb{R}^{q_i \times n}$ is an unknown parameter matrix, and $\varphi_{i,j}(\bar{x}_{i,j}) \in \mathbb{R}^{q_i}$ is a known smooth function.

Agent i is associated with a parameter $r_i > 0$, which plays the role of *safety radius*, by considering its own size and shape. Since the heterogeneous agents are considered here, e.g., they may have different sizes or shapes, individual values for the safety radius may be associated to each agent. Further, it is here assumed that the size and shape of each agent remain fixed and time-invariant. We also denote by $R(\gg r_i)$ the radius of the region in which agents can detect the presence of the other robots (or obstacles).

Suppose that there are $M \in \mathbb{N}$ static obstacles and let $\mu_j \in \mathbb{R}^p$ and $\mathcal{D}_l \subset \mathbb{R}^{p \times p}$ denote the center of mass of the l th ($l = 1, \dots, M$) obstacle and region of the n -dimensional Euclidean space that it occupies, respectively. In what follows elliptical obstacles are considered, i.e.,

$$\mathcal{D}_l = \{q \in \mathbb{R}^p : \|q - \mu_l\|_{D_l}^2 \leq \rho_l^2\}$$

where $0 < \rho_l (\ll R)$ and $D_l = D_l^T > 0$. Further, the standard notation $\partial \mathcal{D}_l$ is employed to denote the boundary of the region

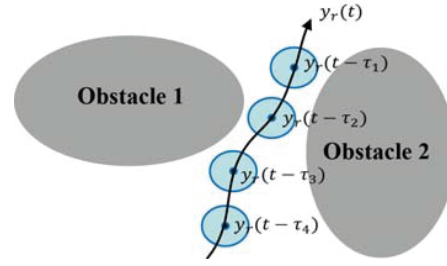


Fig. 1. An example for illustrating Assumption 4. Four agents successively go bypass the obstacle areas along the target trajectory without collisions with each other.

$\mathcal{D}_l$. In practice, an obstacle is typically not elliptical. However, it is possible to enclose the obstacle within a known ellipse by some detection and image processing techniques [27].

From the physical security concerns, each agent should remain a safe distance from the obstacles and the other agents. MAS (1) is said to be *collisions-free* if $\|y_i(t) - \mu_l\|_{D_l} > \rho_l$ and $\|\Delta y_{ij}(t)\| > r_{ij}$ for all $t \geq 0$, $i \neq j \in [N]$ and $j \in [M]$, where $\Delta y_{ij}(t) = y_i(t) - y_j(t)$ and $r_{ij} = \max\{r_i, r_j\}$.

In the MAS, suppose that agent i follows a given target trajectory $y_r(t) \in \mathbb{R}^n$ with a predefined shape $\mathcal{Q}$ (such as desired formation of multiple mobile robots [2], attitude of multiple rigid bodies and so on), where the predefined shape can be formulated as $\mathcal{Q} = (y_1^0, \dots, y_N^0)$, and y_i^0 is the relative location vector and privately known for agent i . More specifically, the output of each agent i asymptotically tracks the preset trajectory $y_i^*(t) := y_r(t) + y_i^0$, i.e., $\lim_{t \rightarrow \infty} \|y_i(t) - y_i^*(t)\| = 0$. It is here assumed that sufficient time is provided to accomplish the tracking task [21]. We attribute this objective to the following problem.

Problem 1 (Collision/Obstacle Avoidance Formation Tracking, COAFT). The collisions-free formation tracking problem consists of finding control inputs $u_i(t)$ that steer all the agents (1) to asymptotically track a given reference $y_r(t)$ with shape $\mathcal{Q} = (y_1^0, \dots, y_N^0)$ while remaining collisions-free, i.e., $\lim_{t \rightarrow \infty} \|y_i(t) - y_i^*(t)\| = 0$, $\|y_i(t) - \mu_l\|_{D_l} > \rho_l$ and $\|\Delta y_{ij}(t)\| > r_{ij}$ for all $t \geq 0$, $i \neq j \in [N]$ and $l \in [M]$.

Assumption 1 [21]. The initial positions and relative location vectors of the agents satisfy $\|\Delta y_{ij}(0)\| > r_{ij}$ and $\|y_i^0 - y_j^0\| > r_{ij}$ for all $i \neq j \in [N]$, respectively.

Assumption 2 [21]. The initial position of each agent and target trajectory do not cross the obstacle region, i.e., $\|y_i(0) - \mu_l\|_{D_l} > \rho_l$ and $\|y_r(t) - \mu_l\|_{D_l} > \rho_l$ for all $i \in [N]$ and $l \in [M]$.

Assumption 3. There exists a time instant T such that $\|y_r(t) - \mu_l\|_{D_l} > R$ for any $t \geq T$ and $l \in [M]$.

Assumption 4. The target trajectory $y_r(t) \in \mathbb{R}^n$ is sufficiently smooth and there exists a positive time-delay value sequence $\{\tau_1, \dots, \tau_N\}$ such that $\|y_r(t - \tau_i) - y_r(t - \tau_j)\| > r_{ij}$ for any $t \geq \max\{\tau_i, \tau_j\}$.

Assumptions 1–3 are fundamental to guarantee the existence

of solution to Problem 1. Assumptions 1 and 2 ensure agents to remain safety distances with each other or external obstacles at initial and target positions. Assumption 3 means that the target trajectory will finally be away from these M obstacles. Assumption 4 ensures that when only a single agent is allowed to pass through the obstacle, each agent can successively go bypass the obstacle areas along the target trajectory without collisions with each other. This phenomenon is illustrated in Fig. 1. In fact, the existence of sequence $\{\tau_1, \dots, \tau_N\}$ can always be ensured if there exists a constant $\underline{y} > 0$ such that $\|\dot{y}_r(t)\| \geq \underline{y}$.

B. Barrier Functions

In this paper, establishing the collisions-free safety constraint can be done through the use of a barrier function $B: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}$ with the following properties: (i) $B(d, x) > 0$, (ii) $B(d, x) < +\infty$, $\|x\| > d$, (iii) $\lim_{\|x\| \rightarrow d^+} B(d, x) = +\infty$. Following the above three properties, we construct the specific barrier function

$$B_D(y, r, R) = \begin{cases} 0, & \text{if } \|y\|_D \geq R \\ +\infty, & \text{if } \|y\|_D \leq r \\ \eta \left(1 + \cos \pi \frac{\|y\|_D^2 - r^2}{R^2 - r^2}\right)^2, & \text{otherwise} \end{cases} \quad (2)$$

where $\eta > 0$ is a design parameter, and D is a positive-definite matrix. Further, we define

$$B_i^a(y_{\mathcal{N}_i}) = \sum_{j \in \mathcal{N}_i} B_I(\Delta y_{ij}, r_{ij}, R_1) \\ B_i^o(y_i) = \sum_{l \in [M]} B_{D_l}(y_i - \mu_l, \rho_l, R_2)$$

where $\max_{i,j \in [N]} r_{ij} < R_1 \leq R$ and $\max_{l \in [N]} \rho_l < R_2 \leq R$ are two design parameters which determine the coverage of B_i^a and B_i^o , respectively; $\mathcal{N}_i := \{j \in [N] : |\Delta y_{ij}| < R\}$.

Obviously, the collisions between agents and obstacles can be avoided if $B_i^a(y_{\mathcal{N}_i})$ and $B_i^o(y_i)$ are bounded, respectively.

III. FORMATION TRACKING CONTROLLER DESIGN

In this section, we give a specific controller design procedure which consists of trajectory planning and trajectory tracking.

A. Trajectory Planning

The following theorem gives the design of a collisions-free reference trajectory method.

Theorem 1. Given a time-delay value sequence $\{\tau_1, \dots, \tau_N\}$ such that Assumption 4 holds, the following reference trajectory

$$y_i^r(t) = \frac{y_i^*(t)}{B_{i,1}^o(B_{i,1}^a + 1) + 1} + \frac{B_{i,1}^o(B_{i,1}^a + 1)y_r(t - \tau_i)}{B_{i,1}^o(B_{i,1}^a + 1) + 1} \quad (3)$$

is collisions-free, where $B_{i,1}^a$ and $B_{i,1}^o$ are the output of the following n_i -order filters of

$$\dot{B}_{i,j}^a = \delta(B_{i,j+1}^a - B_{i,j}^a), \quad j = 1, \dots, n_i - 1 \\ \dot{B}_{i,n_i}^a = \delta(B_i^a(y_{\mathcal{N}_i}) - B_{i,n_i}^a) \quad (4)$$

and

$$\dot{B}_{i,j}^o = \delta(B_{i,j+1}^o - B_{i,j}^o), \quad j = 1, \dots, n_i - 1 \\ \dot{B}_{i,n_i}^o = \delta(B_i^o(y_i) - B_{i,n_i}^o) \quad (5)$$

with $B_{i,j}^a(0) = B_{i,j}^o(0) = 0$, $1 \leq j \leq n_i$, and δ is a positive constant. Also, $\lim_{t \rightarrow \infty} \|y_i^r(t) - y_i^*(t)\| = 0$.

Proof. We consider the following two cases:

Case i) Before entering the coverage of barrier function $B_i^o(y_i)$. Under this case, $\|y_i(t) - \mu_l\|_{D_l} > R_2 > \rho_l$ and $B_i^o(y_i) = 0$. Solving (5) gives that $B_{i,1}^o(t) = 0$ and $y_i^r(t) = y_i^*(t)$. It follows from Assumption 1 that $\|y_i^r(t) - y_j^*(t)\| > r_{ij}$, which implies that $y_i^r(t)$ is collisions-free.

Case ii) After entering the coverage of barrier function $B_i^o(y_i)$. From (5), one has $B_{i,1}^o(t) > 0$.

First, we prove that agent i following trajectory (3) does not collide with the other agents, i.e., $\|y_i(t) - y_j(t)\| > r_{ij}$ for any $t \geq 0$, by reduction to absurdity. For the sake of contradiction, we assume there exist a time t^* and agents i^* and j^* such that $\|y_{i^*}^*(t^*) - y_{j^*}^*(t^*)\| = r_{i^*j^*}$. Since $r_{i^*j^*} < R_1$, then there exists a constant ζ such that $1 + \cos \pi \frac{\|y_{i^*}^*(t^*) - y_{j^*}^*(t^*)\|^2}{R_1^2 - r_{i^*j^*}^2} \geq \zeta > 0$. Then,

$$\begin{aligned} B_{i^*,1}^a &= \int_{\tau=0}^{t^*} C e^{A(t-\tau)} B_{i^*}^a(y_{\mathcal{N}_{i^*}}(\tau)) d\tau \\ &\geq \int_{\tau=0}^{t^*} \frac{(t^* - \tau)^{n_{i^*}}}{n_{i^*}!} e^{-\delta(t^* - \tau)} B_{i^*}^a(y_{\mathcal{N}_{i^*}}(\tau)) d\tau \\ &\geq \frac{h_{i^*}}{n_{i^*}!} \int_{\tau=t^* - \varepsilon}^{t^*} \frac{(t^* - \tau)^{n_{i^*}}}{(\|\Delta y_{i^*j^*}(\tau)\|^2 - \|\Delta y_{i^*j^*}(t^*)\|^2)^{n_{i^*}+1}} d\tau \\ &\stackrel{(a)}{=} \frac{h_{i^*}}{2^{n_{i^*}+1} n_{i^*}!} \int_{\tau=t^* - \varepsilon}^{t^*} \frac{(t^* - \tau)^{n_{i^*}}}{\|\Delta y_{i^*j^*}^T(t') \Delta y_{i^*j^*}(t')\| (t^* - \tau)^{n_{i^*}+1}} d\tau \\ &\geq \frac{h_{i^*}}{2^{n_{i^*}+1} n_{i^*}! \Phi_{i^*j^*}} \int_{\tau=t^* - \varepsilon}^{t^*} \frac{1}{t^* - \tau} d\tau \\ &= +\infty \end{aligned} \quad (6)$$

where

$$A = \begin{bmatrix} -\delta & \delta & & & \\ & -\delta & \delta & & \\ & & \ddots & \ddots & \\ & & & -\delta & \delta \\ & & & & -\delta \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}^T,$$

Eq. (a) follows from the mean value theorem with $t' \in [t^* - \varepsilon, t^*]$, $\Phi_{i^*j^*} = \max_{t \in [t^* - \varepsilon, t^*]} \|\Delta y_{i^*j^*}^T(t) \Delta y_{i^*j^*}(t)\|$, $\varepsilon \in (0, t^*)$ and $h_{i^*} = \eta \zeta^2 e^{-\delta \varepsilon}$. On the other hand, from (5) there exists a constant $\epsilon > 0$ such that $B_{i,1}^o(t^*) > \epsilon$. Thus, (3) becomes $y_i^r(t^*) = y_r(t^* - \tau_i)$. It follows from Assumption 4 that $\|y_i^*(t^*) - y_j^*(t^*)\| > r_{ij}$, a contradiction, which proves that $\|y_i^r(t) - y_j^r(t)\| > r_{ij}$.

Next, we prove that agent i following trajectory (3) does not collide with the obstacles all the time, i.e., $\|y_i^r(t) - \mu_l\|_{D_l} > \rho_l$ for any $t \geq 0$. For the sake of contradiction, we assume that there exist a time t^* , an agent i^* and an obstacle l^* such that $\|y_{i^*}^r(t^*) - \mu_{l^*}\|_{D_{l^*}} = \rho_{l^*}$. Using similar derivations as

(6), one has $B_{i^*,1}^o(y_{i^*}(t^{**})) = +\infty$. According to (3), one has $y_{i^*}^r(t^{**}) = y_r(t^{**} - \tau_{i^*})$. On the other hand, from Assumption 2, $\|y_r(t) - \mu_{l^*}\|_{D_{l^*}} > \rho_{l^*}$ for any $t \geq 0$, a contradiction, which in turn proves that $\|y_i^r(t) - \mu_l\|_{D_l} > \rho_l$ for any $i \in [N]$ and $l \in [M]$.

Summarizing the above two cases, it is shown that reference trajectory $y_i^r(t)$ is collisions-free. Then, $B_{i,1}^a(t) < +\infty$. From (5) and Assumption 3, one has $\lim_{t \rightarrow \infty} B_{i,1}^o(t) = 0$ and $\lim_{t \rightarrow \infty} \|y_i^r(t) - y_i^*(t)\| = 0$. ■

Remark 1. The proposed trajectory planning method highly agrees with practical considerations. In practice, multiple vehicles usually move following a reference route with a preset formation shape, and only after entering the sensing area of the obstacle, the vehicle will reconfigure its formation shape in order to bypass the obstacle. Note that Eq. (5) can be rewritten as

$$y_i^r(t) = \gamma_i(t)y_i^*(t) + (1 - \gamma_i(t))y_r(t - \tau_i)$$

where $\gamma_i(t) \in (0, 1]$ which depends on $B_{i,1}^o$ and $B_{i,1}^a$. This means that the trajectory $y_i^r(t)$ always remains within the generated convex combinatorial space of vectors $y_i^*(t)$ and $y_r(t - \tau_i)$ which are collisions-free. When the MAS is far away from the obstacles ($\gamma_i \rightarrow 1$), all the agents will keep the preset formation shape, and when the MAS is close to the obstacles ($\gamma_i \rightarrow 0$), all the agents will draw close to the target trajectory. Especially, when only a single agent is allowed to pass through the obstacle, each agent can successively go bypass the obstacle areas along the target trajectory without collisions with each other, as illustrated in Fig. 1. The barrier functions $B_{i,1}^o$ and $B_{i,1}^a$ adaptively adjust coefficient $\gamma_i(t)$ to avoid collisions.

Remark 2. Note that the high-order filters (4) and (5) are introduced to avoid involving other state variables $x_{j,2}, x_{j,3}, \dots, x_{j,n_j}$ of the neighboring agent $j \in \mathcal{N}_i$ in repeatedly differentiating reference trajectories (which are unavailable for agent i) in the later backstepping design.

B. Trajectory Tracking

The objective of trajectory tracking is to take the planned trajectory $y_i^r(t)$ as reference signal and construct a distributed tracking controller such that $\lim_{t \rightarrow \infty} \|y_i(t) - y_i^r(t)\| = 0$. Alike, to avoid involving other state variables $x_{i,2}, x_{i,3}, \dots, x_{i,n_i}$ of the neighboring agent $j \in \mathcal{N}_i$ in repeatedly differentiating instruction signals, $B_{i,1}(y_{\mathcal{N}_i})$ is derived by the following n_i -th-order filter

$$\dot{B}_{i,j}(y_{\mathcal{N}_i}) = B_{i,j+1}(y_{\mathcal{N}_i}), \quad j = 1, \dots, n_i - 1 \quad (7)$$

$$\dot{B}_{i,n_i}(y_{\mathcal{N}_i}) = B_{i,n_i}^a(y_{\mathcal{N}_i}) + B_{i,n_i}^o(y_i) \quad (8)$$

Following the developed reference trajectory (3), we use the backstepping technique to develop a distributed adaptive controller to solve the COAFT problem. Specifically, the backstepping design procedure can be presented for agent i based on the following common changes of coordinates

$$z_{i,1} = y_i - y_i^r, \quad (9)$$

$$z_{i,j} = x_{i,j} - \alpha_{i,j-1}, \quad j = 2, \dots, n_i \quad (10)$$

where $\alpha_{i,j-1}$ denotes the virtual control.

We design the virtual control $\alpha_{i,j}$ as follows:

$$\alpha_{i,1} = -c_{i,1}z_{i,1} - \xi_{i,1} + \underbrace{\dot{y}_i^r - \frac{1}{2}B_{i,1}(y_{\mathcal{N}_i})z_{i,1}}_{\text{collision avoidance term}} \quad (11)$$

$$\alpha_{i,2} = -c_{i,2}z_{i,2} - \xi_{i,2} + \Lambda_{i,2} - z_{i,1} \underbrace{-B_{i,1}(y_{\mathcal{N}_i})z_{i,1}}_{\text{collision avoidance term}} \quad (12)$$

$$\alpha_{i,j} = -c_{i,j}z_{i,j} - \xi_{i,j} + \Lambda_{i,j} - z_{i,j-1}, \quad j = 3, \dots, n_i \quad (13)$$

where

$$\Lambda_{i,j} = \sum_{k=1}^{j-1} \frac{\partial \alpha_{i,j-1}}{\partial x_{i,k}} x_{i,k+1} + \sum_{k=1}^{j-1} \frac{\partial \alpha_{i,j}}{\partial y_i^{r(k)}} y_i^{r(k+1)} + \underbrace{\sum_{k=1}^{j-1} \frac{\partial \alpha_{i,j}}{\partial B_{i,k}} B_{i,k+1}(y_{\mathcal{N}_i})}_{\text{collision avoidance term}}$$

$$\varpi_{i,j} = \varphi_{i,j} - \sum_{k=1}^{i-1} \frac{\partial \alpha_{i,j-1}}{\partial x_{i,k}} \varphi_{i,k},$$

$$\tau_{i,1} = \varphi_{i,1}z_{i,1}, \quad \tau_{i,j} = \tau_{i,j-1} + \varpi_{i,j}z_{i,j}, \quad j = 2, \dots, n_i,$$

$$\xi_{i,1} = \hat{\theta}_i^T \varphi_{i,1}, \quad \xi_{i,2} = \hat{\theta}_i^T \varpi_{i,2} - \frac{\partial \alpha_{i,1}}{\partial \hat{\theta}_i} \Gamma \tau_{i,2},$$

$$\xi_{i,j} = \hat{\theta}_i^T \varpi_{i,j} - \frac{\partial \alpha_{i,j-1}}{\partial \hat{\theta}_i} \Gamma_i \tau_{i,j} - \sum_{k=2}^{j-1} \frac{\partial \alpha_{i,k-1}}{\partial \hat{\theta}_i} \Gamma_i \varpi_{i,j} z_{i,k}$$

and the control law and parameter update law for the i th agent are designed as

$$u_i = -c_{i,n_i}z_{i,n_i} - \xi_{i,n_i} + \Lambda_{i,n_i} - z_{i,n_i-1} \quad (14)$$

$$\dot{\hat{\theta}}_i = \Gamma_i \tau_{i,n_i} \quad (15)$$

where $\hat{\theta}_i$ is the estimate of θ_i , Γ_i is a positive definite matrix and $c_{i,j}$ is a positive constant.

From the controller structure (7)–(15), it can be seen that for high-order nonlinear strict-feedback dynamics (1), the proposed controller can be rewritten as the form

$$u_i = \mathcal{U}_i(x_i, y_i^r) + \mathcal{U}_i^{ca}(B_i^o(y_i), B_i^a(y_{\mathcal{N}_i}), z_i)$$

where $\mathcal{U}_i(x_i, y_i^r)$ represents the traditional adaptive backstepping controller given in [28] and $\mathcal{U}_i^{ca}(B_i^o(y_i), B_i^a(y_{\mathcal{N}_i}), z_i)$ with $z_i := [z_{i,1}, \dots, z_{i,n_i}]^T$ represents additional collision avoidance term and $\mathcal{U}_i^{ca}(0, 0, z_i) = 0$. It means that the safety-guarantee controller will reduce to the traditional one when agents keep a safety distance with each other and external obstacles.

The result of this section is now summarized as follows.

Theorem 2. Consider MAS (1) and suppose that Assumptions 1–3 hold. The COAFT problem is solved by the adaptive controller (7)–(15) following reference trajectory (4).

Proof. We construct the integral-type BLF

$$V_i = \left(\int_{\tau=0}^t B_{i,1}(y_{\mathcal{N}_i}(\tau)) d\tau + 1 \right) \frac{\|z_{i,1}\|^2}{2} + \sum_{j=2}^{n_i} \frac{\|z_{i,j}\|^2}{2} + \frac{1}{2} \text{tr} \left\{ \tilde{\theta}_i^T \Gamma_i^{-1} \tilde{\theta}_i \right\}$$

where $\tilde{\theta}_i = \theta_i - \hat{\theta}_i$.

Following the standard backstepping procedure [28], it is not difficult to obtain that

$$\begin{aligned} \dot{V}_i \leq & -c_{i,1} \left(\int_{\tau=0}^t \mathcal{B}_{i,1}(y_{N_i}(\tau)) d\tau + 1 \right) \|z_{i,1}\|^2 \\ & - \sum_{j=2}^{n_i} c_{i,j} \|z_{i,j}\|^2 \end{aligned} \quad (16)$$

Integrating t from $t = 0$ to $+\infty$ for both sides of (16) yields $\sum_{j=1}^N c_{i,j} \int_{t=0}^{\infty} \|z_{i,j}(t)\|^2 dt \leq V_{c,i}(0)$, i.e., $z_{i,j}(t), V_i(t) \in L_2[0, \infty)$. Thus, by using the stability analysis of traditional backstepping controller, all the signals are bounded. Since $z_{i,j}(t) \in L_2[0, \infty)$ and $z_{i,j}(t) \in L_{\infty}[0, \infty)$, by using Barbalat's lemma [28], $\lim_{t \rightarrow +\infty} z_{i,j}(t) = 0$, which in turn implies $\lim_{t \rightarrow +\infty} \|y_i(t) - y_i^r(t)\| = 0$, $i = 1, \dots, N$.

Eq. (16) gives $V_i(t) \leq V_i(0) < +\infty$ for any t and based on this we show $\|y_i(t) - \mu_l\|_{D_l} > \rho_l$ for any $l \in [M]$. For the sake of contradiction, we assume that there exist a time t^* , an agent i^* and an obstacle $\mathcal{D}_{l^*}(t)$ such that $\|y_{i^*}(t) - \mu_{l^*}\|_{D_{l^*}} > \rho_{l^*}$ and $\|y_{i^*}(t) - \mu_{l^*}\|_{D_{l^*}} = \rho_{l^*}$ for $t < t^*$. Under this case, there must exist a constant $\xi_1 > 0$ such that

$$\sum_{i=1}^N \frac{1}{2} \|z_{i,1}(t^*)\|^2 \geq \xi_1 > 0 \quad (17)$$

which can be proved by contradiction. If $\|z_{i,1}(t^*)\|^2 = 0$, then $y_i(t^*) = y_i^r(t^*)$ for $i = 1, \dots, N$. Resultantly, $\|y_{i^*}^r(t) - \mu_{l^*}\|_{D_{l^*}} = \rho_{l^*}$, which contradicts with Theorem 1, and this in turn proves (17).

According to the continuity of $y_{i^*}(t)$, there exists a time t_0^* such that $\rho_{j^*} < \|y_{i^*}(t) - \mu_{l^*}\|_{D_{l^*}} \leq R_2$ for all $t \in (t_0^*, t^*]$. Then, there exists a constant ξ_2 such that

$$1 + \cos \pi \frac{\|y_{i^*}(t) - \mu_{l^*}\|_{D_{l^*}}^2 - \rho_{l^*}^2}{R_2^2 - \rho_{l^*}^2} \geq \xi_2 > 0.$$

Using (17), we obtain [(18) placed at the top of the next page] where $\Psi_n^{i^*l^*} = \max_{t \in [\tau, t^*]} \|\tilde{y}_{i^*l^*}^T(t) D_{l^*} \tilde{y}_{i^*l^*}(t)\|^{n+1}$ with $\tilde{y}_{i^*l^*} = y_{i^*} - \mu_{l^*}$, which is bounded.

Note that Eq. (18) contradicts with $V_i(t) < +\infty$ for all $t \geq 0$, and thus it in turn proves that $\|y_i(t) - \mu_l\|_{D_l} > \rho_l$ for any $l \in [N]$ and $t \geq 0$.

Following the same analysis, it is not difficult to derive $\|\Delta y_{ij}(t)\| > r_{ij}$, $i \neq j \in [N]$ for any $t \geq 0$. ■

Remark 3. Two different types of high-order filters in (4), (5) and (7), (8) are adopted. The filters in (4), (5) may be more complex due to the use of feedback mechanism. Its role is to ensure that $\mathcal{B}_{i,1}^a$ and $\mathcal{B}_{i,1}^o$ can converge to zero when the MAS is far away from the obstacles. Thus, $y_r(t)$ in (3) can converge to $y^*(t)$.

IV. SIMULATION: WITH APPLICATION TO EULER-LAGRANGIAN DYNAMICS

In this section, we apply our control scheme to the problem of formation control of four Euler-Lagrangian agents moving

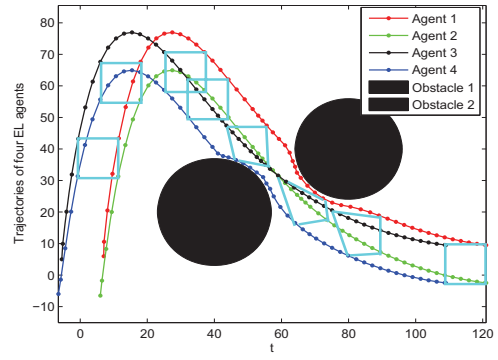


Fig. 2. Dynamic formation reconfiguration of EL multi-agents.

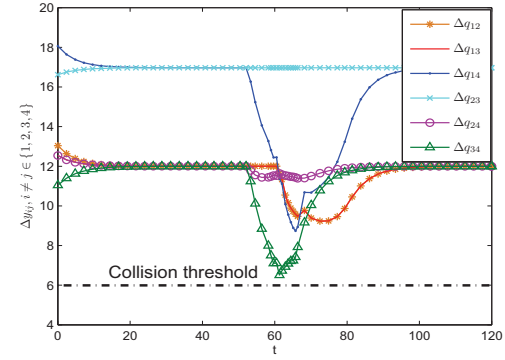


Fig. 3. Relative positions between two EL agents.

on 2-dimensional Euclidean space. The dynamics of agent i is described as follows:

$$M_i(y_i) \ddot{y}_i + C_i(y_i, \dot{y}_i) \dot{y}_i + G_i(y_i) = \tau_i \quad (19)$$

with

$$\begin{aligned} M_i(y_i) &= \begin{bmatrix} b_{i1} + 2b_{i2} \cos y_{iy} & b_{i3} + b_{i2} \cos y_{iy} \\ b_{i3} + b_{i2} \cos y_{iy} & b_{i3} \end{bmatrix} \\ C_i(y_i, \dot{y}_i) &= \begin{bmatrix} -b_{i2} \sin y_{iy} \dot{y}_{iy} & -b_{i2} \sin y_{iy} (\dot{y}_{ix} + \dot{y}_{iy}) \\ b_{i2} \sin y_{iy} \dot{y}_{ix} & 0 \end{bmatrix} \\ G_i(y_i) &= [0, 0]^T, \quad \varphi_i = [b_{i1}, b_{i2}, b_{i3}]^T \end{aligned}$$

where $y_i, \dot{y}_i \in \mathbb{R}^2$ denote the generalized position and velocity vectors, respectively; $M_i(y_i) \in \mathbb{R}^{2 \times 2}$ is the inertia matrix; $C_i(y_i, \dot{y}_i) \dot{y}_i \in \mathbb{R}^2$ is the vector for Coriolis and centripetal forces; $G_i(y_i) \in \mathbb{R}^2$ is the gravity vector; φ_i are unknown system parameters and $\tau_i \in \mathbb{R}^2$ is the control force.

In the simulation, for EL parameter, we set $\varphi_1 = [5, 1, 1]^T$, $\varphi_2 = [5, 1, 2]^T$, $\varphi_3 = [6, 1, 1]^T$ and $\varphi_4 = [6, 1, 2]^T$, and $r_i = 6$. We select the predefined shape as the square $\mathcal{Q} = ([6, 6]^T, [6, -6]^T, [-6, 6]^T, [-6, -6]^T)$. We assume that

$$\begin{aligned}
V_i(t^{**}) &\geq \frac{1}{2} \sum_{i=1}^N \|z_{i,1}(t^{**})\|^2 \int_{\tau=t^*}^{t^{**}} \mathcal{B}_{i,1}(y_{\mathcal{N}_i^*}(\tau)) d\tau \\
&\geq \underbrace{\int_{\tau_n=t^*}^{t^{**}} \int_{\tau_{n-1}=0}^{\tau_n} \cdots \int_{\tau_1=0}^{\tau_2} \int_{\tau=0}^{\tau_1} \frac{\eta \xi_1^2 \xi_2^2}{(\|y_{i^*}(t) - \mu_{l^*}\|_{D_{l^*}}^2 - \rho_{l^*}^2)^{n+1}} d\tau d\tau_1 \cdots d\tau_{n-1} d\tau_n}_n \\
&\geq \int_{\tau_n=t^*}^{t^{**}} \int_{\tau_{n-1}=0}^{\tau_n} \cdots \int_{\tau_1=0}^{\tau_2} \int_{\tau=0}^{\tau_1} \frac{\eta \xi_1^2 \xi_2^2}{(\|y_{i^*}(t) - \mu_{l^*}\|_{D_{l^*}}^2 - (\|y_{i^*}(t^{**}) - \mu_{l^*}\|_{D_{l^*}}^2)^{n+1}} d\tau d\tau_1 \cdots d\tau_n \\
&\geq \int_{\tau_n=t^*}^{t^{**}} \int_{\tau_{n-1}=0}^{\tau_n} \cdots \int_{\tau_1=0}^{\tau_2} \int_{\tau=0}^{\tau_1} \frac{\eta \xi_1^2 \xi_2^2}{2^{n+1} \|\tilde{y}_{i^* l^*}^T(t''') D_{l^*} \tilde{y}_{i^* l^*}(t''')(\tau - t^{**})\|^{n+1}} d\tau \\
&\geq \frac{\eta \xi_1^2 \xi_2^2}{2^{n+1} \Psi_n^{i^* j^*}} \int_{\tau_n=t^*}^{t^{**}} \int_{\tau_{n-1}=0}^{\tau_n} \cdots \int_{\tau_1=0}^{\tau_2} \int_{\tau=0}^{\tau_1} \frac{1}{(t^{**} - \tau)^{n+1}} d\tau d\tau_1 \cdots d\tau_n \\
&\geq \frac{\eta \xi_1^2 \xi_2^2}{2^{n+1} \Psi_n^{i^* j^*}} \int_{\tau_n=t^*}^{t^{**}} \int_{\tau_{n-1}=0}^{\tau_n} \cdots \int_{\tau_1=0}^{\tau_2} \left(\frac{1}{n(t^{**} - \tau_1)^n} - \frac{1}{n(t^{**})^n} \right) d\tau_1 \cdots d\tau_n \\
&\geq \frac{\eta \xi_1^2 \xi_2^2}{2^{n+1} \Psi_n^{i^* j^*}} \int_{\tau_n=t^*}^{t^{**}} \cdots \int_{\tau_2=0}^{\tau_3} \left(\frac{1}{n(n-1)(t^{**} - \tau_2)^{n-1}} - \frac{1}{(n-1)(t^{**})^{n-1}} \right) d\tau_2 \cdots d\tau_n \\
&\quad \dots \\
&\geq \frac{\eta \xi_1^2 \xi_2^2}{2^{n+1} \Psi_n^{i^* j^*}} \int_{\tau_n=t^*}^{t^{**}} \left(\frac{1}{n!(t^{**} - \tau_n)} - \frac{1}{t^{**}} \right) d\tau_n \\
&\geq \frac{\eta \xi_1^2 \xi_2^2}{2^{n+1} \Psi_n^{i^* j^*}} \left(\frac{1}{n!} \log(t^{**} - \tau)|_{\tau=t^*}^{t^{**}} - 1 \right) \\
&= +\infty
\end{aligned} \tag{18}$$

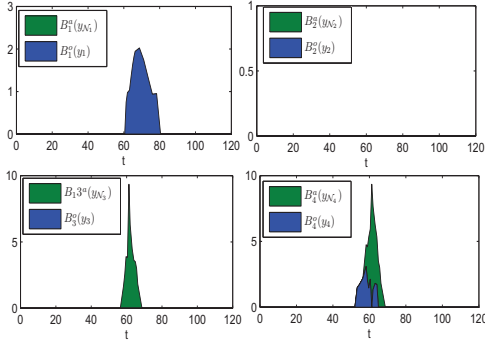


Fig. 4. Barrier functions $B_i^o(y_i)$ and $B_i^a(y_{\mathcal{N}_i})$.

the initial states of the EL systems are

$$\begin{aligned}
[y_1^T(0), \dot{y}_1^T(0)]^T &= [7, 6, 0, 0]^T, \\
[y_2^T(0), \dot{y}_2^T(0)]^T &= [6, -7, 0, 0]^T, \\
[y_3^T(0), \dot{y}_3^T(0)]^T &= [-5.5, 5, 0, 0]^T, \\
[y_4^T(0), \dot{y}_4^T(0)]^T &= [-6.5, -6, 0, 0]^T.
\end{aligned}$$

Obviously, $\|\Delta y_{ij}(0)\| > 6$ and $\|y_i^0 - y_j^0\| > 6$ for all $i \neq j \in [4]$. Hence, Assumption 1 is satisfied.

In the simulation, two circular obstacles are considered, i.e. $\mathcal{D}_1 = \{y \in \mathbb{R}^2 : \|y - [40, 20]^T\|^2 \leq 11^2\}$ and $\mathcal{D}_2 = \{y \in \mathbb{R}^2 : \|y - [80, 40]^T\|^2 \leq 10^2\}$. However, considering the safety radius of agent itself $r_i = 6$, the trajectory of the agent's centroid should remain outside of obstacle regions $\mathcal{D}'_1 = \{y \in \mathbb{R}^2 : \|y - [40, 20]^T\|^2 \leq 17^2\}$ and $\mathcal{D}'_2 = \{y \in \mathbb{R}^2 : \|y - [80, 40]^T\|^2 \leq 16^2\}$. It is easy to verify that Assumption 2 holds.

We select the target trajectory $y_r(t) = [t, 10t \exp(-0.05t) + 0.3 \sin(0.1t)]^T$ and obviously Assumptions 3 and 4 also hold.

Now, we apply the proposed COAFT strategy to complete the formation tracking task. We choose parameters $R_1 = 8$, $R_2 = 18$, $\delta = 1$, $\eta = 30$, $c_{i,1} = c_{i,2} = 0.2$ and $\Gamma_i = I$. Choose an admissible time sequence $\{\tau_1, \tau_2, \tau_3, \tau_4\} = \{0, 1, 2, 3\}$. The simulation results are given in Figs. 2–4. Fig. 2 illustrates that the proposed COAFT strategy can guarantee that these four EL agents adaptively adjust their formation shape to avoid collisions with two obstacles when approaching the obstacles, and after crossing the obstacles, they quickly recovery the predefined shape $\mathcal{Q}$ following the target trajectory $y_r(t)$. Fig. 3 plots the relative distances between any two agents, which are larger than the safety radius all the time. Fig. 4 shows the response curves of barrier functions $B_i^o(y_i)$ and $B_i^a(y_{\mathcal{N}_i})$, which illustrates the effectiveness of these barrier functions on collisions avoidance when the agents approach the obstacles and each other, respectively.

V. CONCLUSION

In this paper, the problem of collision/obstacle avoidance formation tracking was studied for a class of high-order strict-feedback nonlinear multi-agent systems. A dynamic formation reconfiguration method is presented which consisting of a smooth and analytic reference trajectory and adaptive backstepping tracking controller. It is shown that all the agents adaptively reconfigure their formation shape within the barrier function's coverage to avoid collisions with each other and obstacles, while quickly forming a pre-defined shape of formation following the target trajectory outside the barrier function's coverage.

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